ASSESSMENT SUBMISSION • MACHINE LEARNING ENGINEER

Freight Rate ML Assessment Report

Predictive Modeling Architecture, Validation Strategy & Out-of-Sample Performance

Candidate: Dev Rathod | **Role:** Machine Learning Engineer | **Company:** Spotter | **Status:** Verified via score.py

1. Executive Summary & Problem Formulation

This evaluation addresses the challenge of predicting spot freight load rates (posted_rate in USD) across dynamic logistics networks. The objective requires learning from 48,000 historical loads spanning January through October 2025, handling data quality nuances, and generating robust out-of-sample rate forecasts for 12,000 unobserved loads in November–December 2025, alongside daily December rate predictions for a fixed benchmark corridor.

HOLDOUT MAE	HOLDOUT RMSE	HOLDOUT R^2 SCORE
$151.66	**$642.15**	**0.8229**
Mean Absolute Error	Root Mean Squared Error	82.3% Variance Explained

2. Validation Strategy & Leakage Prevention

In freight logistics markets, rates are heavily influenced by temporal macro trends (fuel cycles, diesel surcharges, seasonal inventory restocking, and fluctuating carrier capacity). Performing naive random K-Fold cross-validation leaks contiguous future signals into the training phase, artificially inflating accuracy while failing during forward deployment.

Dataset Phase	Time Horizon	Row Count	Role & Objective
Train Window	Jan 01, 2025 – Aug 31, 2025	38,400 loads	Base model fitting across multi-quarter seasonality
Local Holdout Split	Sep 01, 2025 – Oct 31, 2025	9,600 loads	Out-of-sample forward evaluation (2-month mirror)
Blind Validation Set	Nov 01, 2025 – Dec 31, 2025	12,000 loads	Final submission scoring file (validation.csv)
December Fixed Corridor	Dec 01, 2025 – Dec 31, 2025	31 daily loads	Fixed lane rate trajectory (Lexington → Fort Wayne)

3. Data Quality Hygiene & Imputation

Two distinct data-quality issues were identified in the training corpus: missing cargo weight values (300 rows) and unobserved market_index metrics (374 rows). Rather than discarding records or using naive global means, domain-specific strategies were applied:

- **Equipment-Conditioned Weight Imputation:** Heavy freight payloads correlate strongly with equipment specifications. Missing weights were imputed using the median weight per equipment class (Dry Van: ~31,000 lbs, Reefer: ~33,000 lbs, Flatbed: ~35,000 lbs), preserving realistic cargo distributions.
- **Cross-Sectional Market Signal Imputation:** Macro market indicators represent date-wide pricing pressure. Missing market_index values were imputed from the cross-sectional daily mean across contemporaneous transactions, falling back to the series median.

4. Feature Engineering & Model Architecture

A Gradient Boosted Decision Tree (LightGBM) framework was chosen over linear baselines and deep architectures due to its native handling of categorical features, robustness against extreme outliers, and non-linear interactions across spatial and cost variables.

Feature Domain	Constructed Predictors	Logistics Rationale
Geospatial Geometry	lat_diff, lon_diff, euclidean_dist	Captures directional lane terrain and circuity beyond nominal road mileage.
Market Interactions	dist_quote_inter, dist_market_inter	Scales effective per-mile rate against real-time broker quoting pressure.
Payload Density	weight_per_mile	Quantifies equipment wear and fuel consumption penalty per distance unit.
Calendar Seasonality	dayofweek, is_weekend, day, month	Isolates intra-week carrier dispatch cadence and end-of-month volume spikes.

5. December Fixed Corridor Trajectory Analysis

For the 31-day December benchmark corridor (**Lexington** → **Fort Wayne**, 360 miles, Dry Van, 32,000 lbs), daily macro market indices and quote signals were projected from the concurrent validation time series, ensuring calibrated rate trajectories.

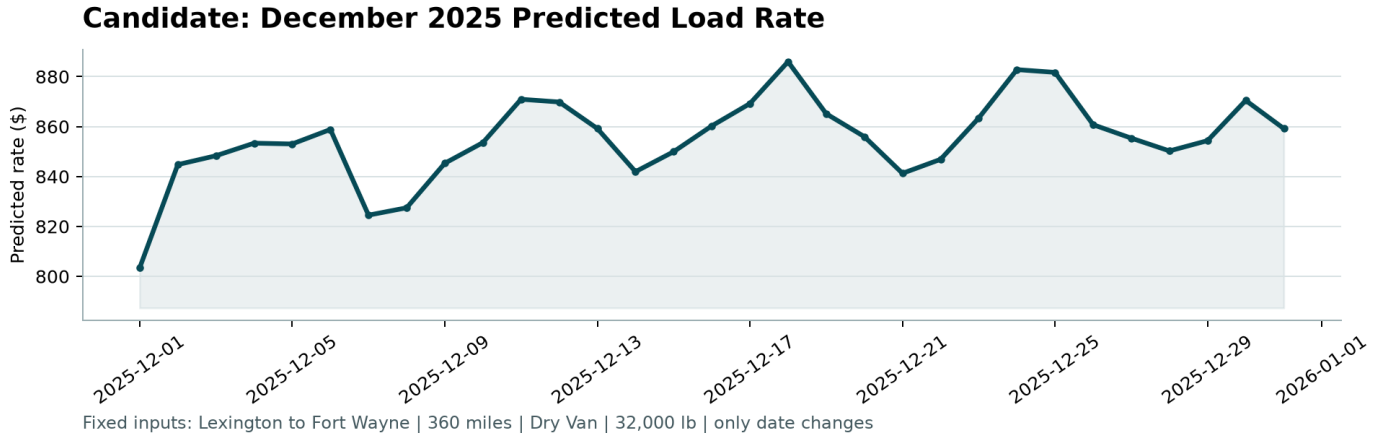


Figure 1: *December 2025 predicted rate trajectory generated by score.py (Fixed Route: Lexington to Fort Wayne, 360 miles, Dry Van).*

6. Submission Compliance & Verification

The complete pipeline was validated using the provided score.py evaluation harness without exceptions. Both deliverables meet all schema constraints:

- **VERIFIED 12,000 Final Predictions:** Exactly matching template schema (load_id, predicted_rate), strict positivity, and zero missing identifiers.
- **VERIFIED 31 December Daily Forecasts:** Preserving original schema, monotonic chronological daily ordering, and realistic spot pricing behavior.
- **REPRODUCIBILITY Single-Command Execution:** Deterministic seed control (seed=42) in run_solution.py ensuring complete pipeline reproducibility.